

Commentary on Wan *et al.*: Optimizing the neurofeedback protocol in addiction treatment

The inconsistent efficacy of EEG-NF is driven by a mechanistic mismatch compounded by methodological heterogeneity. Future research must shift from general efficacy to mechanism-driven protocol design and unified standards (CRED-nf) to identify effective parameters and specific patient subgroups, enabling reliable clinical application.

Neurofeedback (NF) is a form of biofeedback grounded in clinical and behavioral neuroscience that enables individuals to manipulate neural processes by learning to self-regulate brain physiological activity in real-time [1]. Although neurofeedback of brain oscillations based on electroencephalography (EEG-NF) has become the predominant NF modality because of its low-cost and accessibility, its therapeutic utility in drug addiction remains inconsistent [2]. This inconsistency stems from multiple factors, including lack of mechanistic specificity inherent to conventional EEG-NF paradigms to fully engage the dynamic, prefrontal cortex-mediated imbalance that sustains drug-seeking behavior [3]. Another limitation is the small number of well-powered studies available to conclusively establish efficacy and to define optimal NF protocols [4]. The recent systematic review and meta-analysis by Wan and colleagues [5] provides quantitative evidence supporting efficacy of EEG-NF in both substance and behavioral addiction. Importantly, it identifies feedback modality and the number of sessions as primary factors influencing outcomes. The reported findings highlight the need to move beyond demonstrating overall efficacy and instead focus on detailed protocol development and a rigorous mechanistic understanding of which parameters drive therapeutic effects.

Across the trials included in this analysis, outcome measures focused primarily on craving and symptom severity linked to relapse vulnerability. Craving is a core clinical manifestation driven by sensitized mesolimbic dopaminergic signaling and heightened attentional bias, consistent with the incentive sensitization framework [3]. Chronic drug exposure exaggerates cue-driven motivational salience, culminating in hyper-reactive ventral striatal responses [6]. Although most EEG-NF protocols aim to reduce dysregulated processes by training oscillatory patterns linked to arousal or inhibitory control [7], the observed symptom reductions suggest these modulations can influence craving-related neural states. However, the mixed findings point to a critical mismatch: broad target selection does not

adequately capture the specific, dysregulated neural circuits that shape individual relapse profiles.

The meta-analysis identified 'feedback modality' as a key parameter mediating inconsistency. This parameter is often treated as a neutral technical choice, even though the initially observed benefit of using auditory feedback disappears on outlier exclusion. Therefore, a precise mechanistic interpretation necessitates further research and should be considered provisional. Visual feedback, which dominates the literature, risks engaging fronto-striatal reward circuits in unintended ways. For individuals with heightened cue-reactivity, visual stimuli trigger non-specific attentional capture and striatal responsivity [3]. This incidental circuit engagement can compete with the self-regulation task, potentially weakening therapeutic impact [8]. Conversely, non-visual modalities (auditory or haptic) may provide a cleaner neurophysiological channel by minimizing reward-circuit biases. Future protocol design must align sensory delivery channels with known neurocognitive vulnerabilities in addiction phenotypes.

The heterogeneity revealed in this meta-analysis further highlights the urgency of methodological standardization in neurofeedback research. Despite growing interest, the field still lacks consistent reporting of protocol architecture, session dynamics, control conditions and neurophysiological targets. The Consensus on the Reporting and Experimental Design of Clinical and Cognitive-Behavioural Neurofeedback Studies (CRED-nf) provides a robust framework for addressing this gap [9]. Unlike modality-specific guidelines, CRED-nf focuses on principles unique to neurofeedback: signal quality, real-time processing, reward contingencies, training structure and mechanistic justification of the targeted neural markers. Adopting this framework, alongside Consolidated Standards of Reporting Trials (CONSORT) standards, is vital for improving reproducibility and systematically determining which protocol components (e.g. frequency bands, sensory modalities or dose parameters) drive therapeutic change. This methodological rigor is key to advancing EEG-NF from experimental trials to an evidence-based, reliable clinical application in addiction. Taken together, this review positions EEG-NF as a promising neuro-modulation strategy for addiction. However, future impact of EEG-NF depends critically on a paradigm shift toward mechanism-driven protocol design and transparent methodological rigor. Interventions must increasingly incorporate a clear neuroscientific rationale, particularly concerning how protocol choices, such as feedback modality, interact with addiction circuitry. Emerging multi-modal imaging approaches and personalized state monitoring create opportunities for more

adaptive and precision-oriented paradigms in treatment [10]. Moreover, targeted clinical trials are necessary to identify which subgroups (e.g. high cue-reactivity or impulsive/compulsive phenotypes) benefit most from distinct NF strategies. EEG-NF will only achieve reliable clinical utility through mechanistic rigor and methodological unity.

KEYWORDS

addiction treatment, craving, incentive sensitization framework, methodological standardization, neurofeedback, prefrontal cortex

AUTHOR CONTRIBUTIONS

Di Zhao: Conceptualization (equal); writing—original draft (equal). **Björn Apfel:** Conceptualization (equal); writing—review and editing (equal). **Surjo R. Soekadar:** Conceptualization (equal); writing—review and editing (equal).

ACKNOWLEDGEMENTS

D.Z. is supported by National Natural Science Foundation of China grant (82422029, 82271530), the Fundamental Research Funds for the Central Universities (YG2025ZD07), the Science and Technology Commission of Shanghai Municipality (24Y22800200), and China Scholarship Council (202506230324). B.A. and S.R.S. are supported by the German Research Foundation (DFG: Project-ID 402170461-TRR 265), the European Research Council (ERC) under project BNCl2 (101088715) and the Einstein Foundation Berlin, Germany.

FUNDING INFORMATION

National Natural Science Foundation of China, Grant/Award Number: 82422029; Fundamental Research Funds for the Central Universities, Grant/Award Number: YG2025ZD07; Science and Technology Commission of Shanghai Municipality, Grant/Award Number: 24Y22800200; China Scholarship Council, Grant/Award Number: 202506230324; German Research Foundation, Grant/Award Number: 402170461-TRR 265; European Research Council, Grant/Award Number: 101088715; Einstein Foundation Berlin

DECLARATION OF INTERESTS

None.

DATA AVAILABILITY STATEMENT

This is a commentary with no data.

Di Zhao^{1,2} 

Björn Apfel³

Surjo R. Soekadar³

¹School of Nursing, Shanghai Jiao Tong University, Shanghai, China

²Shanghai Key Laboratory of Psychotic Disorders, Brain Health Institute, National Center for Mental Disorders, Shanghai Mental Health Center, Shanghai Jiao Tong University School of Medicine, Shanghai, China

³Clinical Neurotechnology Laboratory, Department for Psychiatry and Neurosciences CCM, Charité-Universitätsmedizin Berlin, Corporate Member of Freie Universität Berlin and Humboldt-Universität zu Berlin, Berlin, Germany

Correspondence

Di Zhao, School of Nursing, Shanghai Jiao Tong University, Shanghai 200025, China.

Email: dizhao_sunset@sjtu.edu.cn

Surjo R. Soekadar, Clinical Neurotechnology Laboratory, Department for Psychiatry and Neurosciences CCM, Charité-Universitätsmedizin Berlin, Corporate Member of Freie Universität Berlin and Humboldt-Universität zu Berlin, Berlin 10117, Germany.

Email: surjo.soekadar@charite.de

ORCID

Di Zhao  <https://orcid.org/0000-0002-6830-8920>

REFERENCES

1. Sitaram R, Ros T, Stoeckel L, Haller S, Scharnowski F, Lewis-Peacock J, et al. Closed-loop brain training: The science of neurofeedback. *Nat Rev Neurosci*. 2017;18(2):86–100. <https://doi.org/10.1038/nrn.2016.164>
2. Dousset C, Kajosch H, Ingels A, Schröder E, Kornreich C, Campanella S. Preventing relapse in alcohol disorder with EEG-neurofeedback as a neuromodulation technique: a review and new insights regarding its application. *Addict Behav*. 2020;106:106391. <https://doi.org/10.1016/j.addbeh.2020.106391>
3. Ceceli AO, Huang Y, Kronberg G, McClain N, King SG, Butelman ER, et al. The impaired response inhibition and salience attribution model of drug addiction: recent neuroimaging evidence and Future Directions. Annual review of the incentive-sensitization theory of addiction 30 years on. *Annu Rev Psychol*. 2025;77. <https://doi.org/10.1146/annurev-psych-040725-025923>
4. Lucas I, Sole-Morata N, Baenas I, Rosinska M, Fernandez-Aranda F, Jimenez-Murcia S. Biofeedback interventions for impulsivity-related processes in addictive disorders. *Curr Addict Rep*. 2023;10(3):543–52. <https://doi.org/10.1007/s40429-023-00499-y>
5. Wan H, Chen B, Li X, Bu J. Effect of EEG neurofeedback therapy on addiction disorders: A systematic review and meta-analysis. *Addiction*. 2025. <https://doi.org/10.1111/add.70164>
6. Stoeckel LE, Weller RE, Cook EW III, Twieg DB, Knowlton RC, Cox JE. Widespread reward-system activation in obese women in response to pictures of high-calorie foods. *Neuroimage*. 2008;41(2):636–47. <https://doi.org/10.1016/j.neuroimage.2008.02.031>
7. Bu J, Young KD, Hong W, Ma R, Song H, Wang Y, et al. Effect of deactivation of activity patterns related to smoking cue reactivity on nicotine addiction. *Brain*. 2019;142(6):1827–41. <https://doi.org/10.1093/brain/awz114>
8. Ros T, Enriquez-Geppert S, Zotev V, Young K D, Wood G, Whitfield-Gabrieli S, & Thibault R T. Consensus on the reporting and experimental design of clinical and cognitive-behavioural neurofeedback studies (CRED-nf checklist). 2020.
9. Krause F, Linden DE, Hermans EJ. Getting stress-related disorders under control: The untapped potential of neurofeedback. *Trends Neurosci*. 2024;47(10):766–76. <https://doi.org/10.1016/j.tins.2024.08.007>